Personal Statement

National University of Singapore
Jianming Xing MSc in Mechanical Engineering

Mechanism geometry and kinematics gave me a language for machines: dimensions constrain motion, and transformations connect joint coordinates to an end effector. Yet a useful robot cannot stop at a correct geometric model. Its cameras must produce interpretable signals, software must exchange coherent states, controllers must convert decisions into stable motion, and learning methods must be evaluated against physical behavior. My work has progressed from analyzing how mechanisms move to asking how perception, software, control, and learning can form one dependable machine.

I built this progression through complementary degrees. In June 2025, I earned a B.Eng. in Mechanical Design, Manufacturing and Automation at Harbin Institute of Technology, Weihai. Its training in mechanism behavior, kinematics, and physical constraints established my foundation in robotic systems. I am now pursuing a second bachelor's degree in Computer Science and Technology at Harbin Institute of Technology, Shenzhen, expected in June 2027. This is a deliberate addition to my mechanical preparation: algorithms, data structures, and software interfaces allow me to construct machines whose capabilities extend beyond fixed geometry. Studying computation alongside mechanics lets me reason from physical design through perception and control to integrated operation.

The Robot Digital-Twin HMI for Weld-Seam Recognition joined those domains in one engineering system. For its LeArm representation, I built a modified-DH model and derived analytical forward and inverse kinematics. I used the same transformation chain to map joint inputs into end-effector and link poses in an OpenGL digital twin assembled from STL models. My engineering decision was to treat geometry as a shared interface, keeping numerical kinematics, joint controls, and the visual state under one coordinate convention. I also integrated C++/Qt controls with Python/PyTorch segmentation inference while preserving explicit boundaries among desktop interaction, inference, and robot visualization. The original U-Net recorded 81.19% mIoU and 0.64 seam-class IoU, while the final 939-image configuration recorded 96.80% mIoU and 0.94 seam-class IoU. Because architecture, pretraining, and dataset size changed across the recorded stages, I treated the results as a configuration sequence rather than a controlled single-factor ablation. The perceptual output remained 2D pixel-level segmentation, and the robot moved to a fixed preset pose rather than a trajectory derived from seam geometry. Integration therefore depended on coherent boundaries between geometry, perception, and motion.

In a later project integrating a Dobot CR5, an Orbbec Astra2 RGB-D camera, and an electric gripper, I encountered related constraints. Eye-to-hand calibration yielded a usable static transform, which I used only for fixed-view RGB data collection and inference; I treated residual reprojection error as an explicit operating constraint because high-precision calibration remained incomplete. I translated physical limits into workspace-bound, start-pose-distance, and per-step motion checks, followed by slow playback before using or replaying data. In the ROS2 real-robot loop, ServoP drove the arm and Modbus controlled the gripper. Non-blocking execution could let a new inference use an intermediate in-motion state before the preceding action chunk had finished, producing back-and-forth corrections. I therefore selected blocking, short-horizon ServoP chunks, accepting lower apparent frequency to retain state-action consistency and trajectory quality. Workspace validity and execution consistency thus became separate checks: the first constrained admissible targets, while the second constrained when a new observation could meaningfully drive the next command. I evaluated the machine through reference frames, admissible motion, timing, and the execution schedule rather than component performance alone.

My Xbotics Embodied AI Community Internship provided another test of systems integration. I migrated Isaac Lab's ANYmal-C navigation task into a NumPy-based MotrixLab environment, refactoring reset and step flows, command sampling, observations, rewards, and termination. I treated the migration as preservation of behavior-defining interfaces rather than translation of code alone. After correcting MuJoCo heightfield frames and constraining spawn and goal locations, I added rollout and reward/termination regression checks. I used them to check how changes affected coordinate handling, state assembly, reward calculation, and termination behavior. My engineering judgment was to make such assumptions testable at the boundaries among mechanics, simulation, and learning.

My immediate goal is an R&D engineering role in robotics, automation, or intelligent manufacturing, connecting mechanics with perception, control, and dependable software. I want to deepen my ability to model physical systems, evaluate data-driven methods, and validate their behavior on real hardware. After gaining stronger real-system experience, I may pursue doctoral study; it remains an option rather than a predetermined route. I therefore seek program-specific training and applied opportunities that can turn this cross-disciplinary foundation into rigorous machine development.

In my weld-seam HMI, modified-DH kinematics linked joint controls to an OpenGL digital twin; in the separate CR5 project, workspace checks and execution timing connected software decisions to physical motion. I can integrate these layers, but I need a more systematic pathway from robot mechanics through computer control and mechatronics to project and industry-based validation. NUS's MSc in Mechanical Engineering would help me develop that pathway for intelligent machines.

If the current offerings remain available in my intake, ME5402 Advanced Robotics and ME5422 Computer Control and Applications would strengthen the robotics and control foundations beneath my integration decisions. Subject to availability and approval, the optional four-unit ME5001A Mechanical Engineering Project and optional four-unit ME5888M Mechanical Engineering Internship could form a four-plus-four combination within the eight-unit cap on project and internship courses. ME5001A could let me investigate a focused Control & Mechatronics or Manufacturing question through measured mechanical and control behavior. ME5888M at a local industrial company could then expose the same method to industrial constraints, where feasibility and integration matter alongside laboratory performance. I would bring experience making coordinate, workspace, and timing assumptions testable across a digital model and real robot. In robotics or intelligent-manufacturing R&D, I intend to develop machines whose mechanical, control, and software behavior can be evaluated together.